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Combined use of web scraping and AI-based models for business applications: research evolution and future trends

Giuliana Barba[1] · **Marianna Lezzi**[1] · **Mariangela Lazoi**[1] · **Angelo Corallo**[1]

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Abstract

Online data are instrumental in knowledge generation and decision-making processes, particularly for companies aiming to glean insights for marketing strategies and service optimizations. Web scraping emerges as a pivotal technique for automatic data collection from the internet, crucial in today's competitive business landscape where online presence is paramount. Coupled with advanced artificial intelligence (AI) models, web scraping becomes even more powerful, enabling deeper analysis of unstructured data for enhanced decision-making. However, managing and analysing the diverse online information presents challenges. Through a bibliometric analysis conducted on 567 collected papers, this study aims to elucidate the evolving business landscape of integrating web scraping with AI-based models. In particular, it addresses key research questions regarding the evolution, business implications, and future trends of this integration. Results show a significant growth in research in this domain, with implications spanning sentiment analysis, natural language processing integration, data intelligence, and deep learning integration. The critical interpretation of the results has allowed for the definition of a future research agenda that proposes concrete directions for the development of more transparent, safe, sustainable and democratic technologies. The paper provides both theoretical and practical contributions. From an academic perspective, it offers insights into the effective utilization of AI-based models and web scraping integration for optimized scraping strategies in business, providing a clear overview of emerging research direction in this field. From a managerial perspective, this paper guides managers in adopting advanced AI-based models to positively impact their company's decision-making involving a strategic and interdisciplinary reflection on the responsible use of scraping and AI.

Keywords Web crawling · Artificial intelligence · Decision support · Text mining · Future research directions · Bibliometric analysis

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Extended author information available on the last page of the article

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1 Introduction

The World Wide Web is the largest global platform for communication and exchanging opinions (Varlamov and Turdakov 2016). It contains a vast amount of heterogeneous data that are easily accessible to the public from various sources, including forums, blogs, microblogs, and social networks (Khder 2021). Online data are a crucial tool for knowledge generation and knowledge-based decision support (Craven et al. 2000). In particular, online data enable companies to consolidate information by transforming it into useful insights for marketing and service decisions (Pantano et al. 2019), such as optimising pricing decisions based on consumer behaviour (Xu et al. 2016) or segmenting customers based on their perceptions (Rejeb et al. 2020). Moreover, companies can harness the web as a valuable resource, tapping into a wealth of information encompassing consumer opinions (Corallo et al. 2020), constructive employee feedback (Tanasescu et al. 2022), and insights pertaining to cutting-edge industrial technologies (Wang et al. 2017). Gathering online data not only facilitates a comprehensive analysis of web users' perceptions of corporate products and services (Varlamov and Turdakov 2016), but also contributes positively to the enrichment of the company's information assets (Diouf et al. 2019).

Web scraping is a valuable technique for automatic data collection from the internet (Tanasescu et al. 2022) and is essential in today's business environment where a company's online presence is vital for remaining competitive (Khder 2021). In particular, this technique is useful for the creation of datasets, such as on third-party products (Thota and Ramez 2021) or customers feedback (Bisconti et al. 2019), and for managing missing or incomplete data, such as reviews of particular products (Miao et al. 2018). However, the heterogeneity of information, such as texts, images or videos, present on the web makes the process of managing and analysing this information challenging (Allahyari et al. 2017).

To enhance the value of these information, advanced artificial intelligence (AI)-based models are used to extract hidden insights from unstructured data (such as news and opinion on web) to enhance decision-making strategies (Tanasescu et al. 2022), to understand consumer behaviour (Bisconti et al. 2019) and the public attitudes towards a company or its products (Xu et al. 2016).

The combine use of web scraping and AI-based models opens up new possibilities in the field of web data extraction and analysis. In particular, AI-based models improve the web scraping process by estimating the relevance of information (Kumar et al. 2021) and adapting more easily to the dynamic structures of web pages (Allahyari et al. 2017). On the other hand, AI models allow to perform more in-depth analyses on unstructured data extracted online (Zhang et al. 2023), such as evaluating the honesty of customer reviews (Sahu et al. 2022) or the opinions of your employees (Tanasescu et al. 2022).

Despite these contributions, remains a significant gap in understanding the integrated implications of web scraping and AI-based models in business contexts. The literature lacks a coherent future research agenda that outlines evolving research trajectories and explores the broader business implications of this integration, going beyond a purely technical or competitive advantage framework.

To date, there is a lack of investigations into how these technologies impact organizational resilience, digital ethics, or systemic business risks. This article therefore aims to investigate systemic implications, including potential cyber vulnerabilities of automated systems, the ethics of data collection from open web sources, and societal implications. By addressing these neglected dimensions, this study contributes to a more holistic understanding of the field and encourages a research agenda that is not only driven by innovation, but also by awareness.

In order to fill this gap, a Bibliometric Analysis, based on a systematic data collection, is conducted with to highlight the main implications of the combined use of AI-based models and web scraping, as well as providing an overview of its temporal evolution and future research trends.

In particular, the paper aims to provide an answer to the following research questions (RQ):

RQ1. How has research into integrating web scraping with AI-based models evolved?
RQ2. What are the business implications of applying web scraping integrated with advanced AI-based models?
RQ3. What are the future research trends related to the joint use of web scraping techniques and AI-based models?

To achieve the objective of the paper and to address the research questions, 567 references related to web scraping and advanced AI models applied to business were retrieved from Scopus and Web of Science (WoS) database.

Firstly, a performance analysis is conducted, based on several basic bibliometric indicators (e.g., number of publications and country), to identify the publications trend over the time, the distribution of document types, the distribution of papers by subject area and a ranking of the most active countries in research in this field.

Secondly, a visual analysis of textual bibliographic data (i.e. title, abstract and keywords) is conducted using VOSviewer application, useful to analyse the keyword co-occurrences extracted from the bibliographic paper data, which allowed the identification of three thematic clusters (i.e., “Intelligent Web Crawling”, “Advanced Sentiment Analysis”, “Digital Social Network") that suggest potential research areas and the business implications of the combined use of web scraping and advanced AI-based models.

Therefore, this study provides both theoretical and practical contributions. On the one hand, it expands the boundaries of knowledge by formulating possible future research directions in the field of AI-based models and web scraping providing a detailed insight into how such models can be effectively used in the business context to improve decision-making processes and optimise integration strategies. On the other hand, it supports managers and practitioners in understanding which innovative AI-based models they could adopt to positively influence business decisions and how to exploit web data obtained by scraping. Moreover, the study offers new opportunities for companies to manage and extract insights from vast amounts of online data, which is particularly significant in overcoming the challenges of managing dirty data and obtaining missing information, thus ensuring reliable business decisions.

The remaining part of the paper is structured as follows: Sect. 2 is dedicated to providing background on web scraping, advanced AI-based models, as well as combinations of such technologies uses in business applications; Sect. 3 describes the methodology adopted in this research; Sect. 4 presents the results of the exploratory study. Finally, Sect. 5 provides a detailed discussion of the business implications and the evolution of research in web scraping and advanced models based on AI, and proposes a future research agenda to address the challenges associated with the impact of these technologies; while, Sect. 6 closes the paper.

2 Theoretical background

2.1 Web scraping for business data collection

Web scraping, also referred to as web crawling (Khder 2021), web harvesting (Zhao 2017), or screen scraping (Arjunan 2022), is a highly efficient technique in the realm of web data gathering (Barbera et al. 2023). It involves systematically extracting unstructured data from websites and transforming it into structured data for storage in a file or database (Khder 2021).

This technique is automated using virtual agents which perform web search actions automatically, emulating human intervention (Diouf et al. 2019; Tanasescu et al. 2022). In particular, web scraping involves the implementation of two main components: the crawler, responsible for downloading data from the web (Khder 2021; B. Zhao 2017), and the scraper, which extracts the desired information from the downloaded data and organizes it according to the structure defined by the user (Khder 2021; Thota and Ramez 2021; B. Zhao 2017).

In the context of data science, web scraping is recognized as an efficient automated tool for collecting large amounts of data (Barbera et al. 2023), ensuring consistently updated information in line with the dynamism of sources (Bhardwaj et al. 2021). Thanks to low execution and maintenance costs (Khder 2021), in addition to great customization capability (Hillen 2019), web scraping allows for greater efficiency in extracting data from various online sources, facilitating the analysis and organization of information for various purposes (Khder 2021). For instance, companies can use web scraping to monitor the change in prices of competing products (Hillen 2019; Jorge et al. 2020), to retrieve missing data and consolidate information from various sources (Singrodia et al. 2019), to collect and investigate employee opinions (Tanasescu et al. 2022) or to predict future skills (Dawson et al. 2020).

Despite the multiple applications and advantages resulting from the use of web scraping in online data collection, various challenges arise that limit its effectiveness. One of the main obstacles is the difficulty in identifying specific URLs containing target information (Arjunan 2022) exacerbated by the irregular and dynamic structure of some web pages (Arjunan 2022). In fact, the search for the right data source, that guarantees relevance and quality, is often a manually executed task, requiring time and cognitive skills (Azad et al. 2018; Diouf et al. 2019). This problem then spills over into the need to evaluate and cleanse the data obtained from web scraping (Tanasescu et al. 2022). Finally, some websites actively seek to prevent web scraping

to avoid service disruptions caused by heavy traffic (Arjunan 2022), prompting many of them to provide a web Application Programming Interface (API) with limits on the number of allowed requests (Wang et al. 2019).

Table 1 provides an overview of the main features of web scraping, highlighting their benefits, limitations and business applications.

2.2 Advanced AI-based models for business applications

Given that online information is predominantly presented in an unstructured format, such as text and multimedia content (Chan et al. 2022), this section offers an overview of AI-based models specialized in processing such data, with a focus on their business applications.

The importance of AI-based models in the business domain is highlighted by their capability to undertake in text documents the Natural Language Processing (NLP), defined as the ability to understand human language (Chan et al. 2022),with proficiency in syntactic and semantic sentence properties (Iqbal and Qureshi 2022). A specific example is the TechNet model (Sarica et al. 2020), which demonstrates how AI

Table 1 Benefits, limitations and application examples of the main features of web scraping for business data collection

Features	Benefits	Limitations	Business applications
Systematic data extraction	Systematic extraction of unstructured data from websites (Thota and Ramez 2021)	Difficulty in identifying specific URLs, non-functional links, or irrelevant sites (Arjunan 2022)	Automated extraction and organization of information according to user-defined structure (Khder 2021)
Automated virtual agent	Automated search and data extraction, eliminating the need for manual operations (Tanasescu et al. 2022)	Risk of being blocked by anti-scraping measures (Arjunan 2022)	Automated downloading and extraction of relevant data from various online sources (Zhao 2017)
User-friendly operation	Easily automated using open source or paid solutions (Arjunan, 2022)	Dynamic and irregular web pages may lead to scraping errors (Varlamov and Turdakov 2016)	Support for users in extracting online data without in-depth programming knowledge (Arjunan, 2022)
Data collection for business analysis	Efficient method with low execution and maintenance costs (Khder 2021)	Need for manual validation and data cleaning rules (Tanasescu et al. 2022)	Collection of large amounts of business-critical data, from competitor prices to customer opinions (Hillen 2019; Jorge et al. 2020)

can be used to extract relational knowledge from complex patent documents. Additionally, this type of model is proving valuable in rapidly improving business taxonomies by exploiting large corpora of textual documents (Arslan and Cruz 2022). An AI-based model for NLP applications has been explored also in the field of human resources to forecast optimal candidates facilitating decision-making (Sridevi and Suganthi 2022).

Moreover, AI-based models are pivotal in audio, video, and image processing (Chan et al. 2022), demonstrating how innovative technologies can be transformed into practical business solutions. For instance, Latham and Giannetti (2023) introduced a framework for hot rolling mill process defect detection using image classification AI-based model. Similarly, Yuan et al. (2020) developed an AI-based model for detecting faults in bearings through the analysis of vibration signal images. Moreover, Tang et al. (2020) demonstrates the significant advancements made in harvesting robots and vision technology research, and its application to the agricultural sector. In addition, Erlina and Fikri (2023) propose a highly effective monitoring system based on AI models that identifies and categorises visits to retail shops, empowering owners to make informed decisions concerning visitor interaction and safety measures. In the same sector, Jafriz and Mansor (2022) propose an innovative system based on AI-models for tracking customers flux in retail establishments. Furthermore, Babu et al. (2020) present a cost-effective solution for tracking people to assist camera operators during filming. Finally, Liu et al. (2020) propose an AI-model-based platform for recording scheduling data, using voice recognition technology, to support professionals in business energy analysis.

Despite numerous benefits and potential applications, the adoption of AI-based models in a business context faces significant challenges. First of all, considering the fluidity of language and the evolution of word meanings in various contexts, the lexicon-based method has limited accuracy in forming a catalogue of previously categorized terms (Corallo et al. 2020). An additional obstacle frequently encountered is the utilization of these technologies as 'black boxes,' leading to indeterminate output outcomes. Companies must achieve a delicate equilibrium between the precision of applied methods and the assessment of outcomes with the use of suitable metrics to measure effectiveness (Lee and Shin 2020). Moreover, the importance of assessing the prolonged efficacy and performance of tools arises, confirming cognisance of the necessity to appraise sustainability in the long run (Latham and Giannetti 2023). Finally, it is common for AI-based models to rely on limited data sets, which creates a gap between the model's understanding and the complexity of the external world (Li et al. 2021).

Table 2 provides an overview of the main features of AI-based models, putting into evidence their benefits, limitations and business applications.

2.3 Business benefits of combining web scraping and advanced AI-based models

The integration of advanced AI-based models with web scraping techniques facilitates the optimization of online data collection, by filtering out redundant information and ensuring the extraction of more pertinent and clean data (Arjunan 2022; Reddy et al. 2021).

Table 2 Benefits, limitations and application examples of the main features of AI-based models for business data analysis

Features	Benefits	Limitations	Business applications
Data heterogeneity processing	Handling large streams of diverse data coming from different sources (Afandizadeh et al. 2023)	Dependency on complete, high-quality data for accurate analysis (Li et al. 2021)	Analysis of online data for identifying behavioural patterns and preferences (Chan et al. 2022)
Unstructured data processing	Processing of unstructured data like text and multimedia (Afandizadeh et al. 2023)	Complexity of natural language grammar presents challenges in NLP applications Liu et al. (2020)	Development of detailed segmentation strategies for personalized marketing campaigns Erlina and Fikri (2023)
Business data analysis	Identification of effective patterns and trends for prediction, segmentation, and classification (Chan et al. 2022)	Balancing precision and outcome evaluation with suitable metrics is a challenge (Lee and Shin 2020)	Informed decision-making and development of precise business strategies by companies (Chan et al. 2022)

This targeted approach benefits businesses by enabling the extraction of essential business information from the web, fostering the acquisition of meaningful data to guide strategic decisions (Lee and Na 2023). For instance, Thuan et al. (2022) combined web scraping and an AI-based model to establish connections between jobs and corporate recruiters, simplifying the process of selecting an ideal job based on skills. Meanwhile, Afandizadeh et al. (2023) introduce a predictive AI-based model for vehicle sales, leveraging specification data from the web to improve dealership sales strategies; while, Wang et al. (2017) suggest that implementing an AI-based model for data acquisition and querying via web scraping can significantly contribute to competitor price comparison and inventory management.

Moreover, AI-based models demonstrate their crucial role in enhancing web scraping accuracy by filtering relevant web pages and information (Arjunan 2022) and summarizing insights from online documents (Reddy et al. 2021), offering valuable assistance for strategic decisions in the business context.

Furthermore, combined use of web scraping and AI-based models plays a pivotal role in furnishing vital insights into consumer sentiment, thereby aiding businesses in making well-informed decisions within the competitive business landscape (Sahu et al. 2022). For instance, Kafeza et al. (2023) leverage social media data to model consumer behaviours, while (Tanasescu et al. 2022) streamline decision-making processes collecting employee feedback online. On the other hand, Sahu et al. (2022) propose an AI-based model to extract reviews from the web and evaluate them for honesty, providing companies with reliable consumer perceptions in the e-commerce market.

The combination of AI-based models and web scraping has potential, but given the early stage of development, further research is needed to improve the performance and accuracy of AI-based models (Bhardwaj et al. 2021; Sahu et al. 2022; Tanasescu et al. 2022). The effectiveness of AI-based models is often hindered by the inconsistent and unstructured nature of data manipulation (Bhardwaj et al. 2021) and requires advancements in data labelling process to obtain more structured and consistent information (Hari Krishna et al. 2021). In addition, models that specialise in sentiment analysis have gaps in the detection of sarcasm and irony that compromise the accuracy of the analysis (Tanasescu et al. 2022). Finally, the integration of human expertise is paramount in interpreting the results of AI-based models, highlighting the importance of collaborative efforts between AI and human insight to improve the applicability and quality of extracted information (Kumar et al. 2021).

The Table 3 illustrates how the integration of AI-based models and web scraping is employed across diverse domains to enhance business optimization strategies.

3 Research methodology

The aim of this study is to examine the evolution of research on AI-based models in combination with web scraping, identify the business implications and highlight future research trends in this area. To achieve this, a systematic approach for data collection is first employed to gather coherent documents from the literature, enhancing the transparency, scientific rigour and replicability of the study (Tranfield et al. 2003).

Secondly, a bibliometric analysis is carried out to examine the collected sample and gain an exploratory understanding of the main themes of interest. Bibliometric analysis, introduced by Pritchard (1969), is recognized as a crucial methodology for exploring research in various disciplines, highlighting its nature, and providing

Table 3 Main optimization strategies for combined use of web scraping and AI-based models

Optimization Strategy	Description	References
Web data extraction	AI-based models optimize the web scraping process by recognizing and categorizing relevant data, focusing on potentially useful websites	Arjunan (2022), Bhardwaj et al. (2021), Kumar et al. (2021), Lee and Na (2023), Reddy et al. (2021), Sahu et al. (2022)
Business analysis	AI-based models for in-depth analysis of business data integrated with web data to identify opportunities for improvement and optimization of operations	Afandizadeh et al. (2023), Hao et al. (2023), Kinne and Resch (2018), Thuan et al. (2022), Wang et al. (2017)
Sentiment/opinion/behavioural data analysis	AI-models detect hidden patterns in online posts, significantly contributing to data collection through web scraping	Kafeza et al. (2023), Khder (2021), Tanasescu et al. (2022)

particularly valuable insights in fields still in development (Donthu et al. 2021a, ; Shao and Shi 2023). The methodology proposed by Noyons (1999) for bibliometric analysis was specifically chosen. This involves preliminary performance analysis and science mapping analysis of the metadata present in the selected papers. Specifically, in the preliminary analysis, several indicators are used (such as the number of publications, geographical distribution, document type and thematic areas). Whereas, the science mapping phase uses the VOSviewer application to analyse the co-occurrences of keywords in the titles and abstracts of the articles with the aim of reducing author bias, understanding the relationships between the topics and their evolution over time (Perianes-Rodriguez et al. 2016).

The entire research process can be summarized into three main steps for data collection and analysis (see Table 4): definition of the search framework and data sample, performance analysis and science mapping analysis.

To support the interpretation of the results emerging from the bibliometric analysis, the collected sample of 567 papers was exported in CSV format and processed using Microsoft Excel, where bibliographic information (e.g., publication year, document type, subject area, title, abstract) was categorized to allow efficient filtering and cross-checking. Selection criteria were also combined where appropriate, for example to track thematic developments in specific time periods or subject domains. The selection of articles occurs jointly with the analysis of bibliometric results, following an iterative and data-driven approach (Donthu et al. 2021a).

This methodological choice is dictated by the need to retrieve, in a targeted and contextualized way, the most relevant studies with respect to the specific factors examined, such as the year of publication, the type of document, the disciplinary area and the keywords that emerged from the analysis. In this way, it is possible to select 145 representative contributions that could integrate and strengthen the interpretation of quantitative results with relevant and up-to-date qualitative evidence (Öztürk et al. 2024).

Table 4 Research methodology

Phase	Activities
1. Research Framework and Data Sample	Utilize Scopus and WoS scientific database to conduct the data sample collection: - Initial sample on Scopus: 606 papers - Initial sample on WoS: 217 papers - Exclusion of notes, dataset and conference proceedings combined with English-language refinement and Duplicate removal: 567 papers
2. Performance Analysis	Utilize MS Excel for a preliminary analysis, examining statistical trends: - Publication and citation count per year - Distribution by document type - Distribution by subject area - Most productive Countries and citation impact
3. Science Mapping Analysis	Perform bibliometric analysis using VOSviewer, including: - Network visualization - Overlay visualization - Density visualization

Specifically, the titles and abstracts of the collected papers were examined by adopting different inclusion criteria based on the specific needs that emerge during the analysis of the bibliometric results as reported in Table 5.

4 Research framework and data sample

To proceed to an accurate analysis, the keywords that fully represent the scope of the work were identified following several steps. First, preliminary research was carried out on Google Scholar on the application of web scraping techniques and models based on AI in corporate contexts. This approach has allowed us to acquire a broader knowledge of these issues and to identify the potential of integrating these tools. Furthermore, based on the consideration of similar concepts, synonyms, and acronyms identified in the theoretical context, it was possible to identify the keywords to use to perform a systematic search (Corallo et al. 2021). The query scheme used for data collection is divided into four groups of keywords (see Table 6).

The first group of keywords helps to identify specific research on the use of web scraping; while, the second group covers technologies and methods associated with AI, which are crucial to understanding how predictive and analytical models can be applied in business activities. Since this research focuses on the application of web scraping and AI-based models in business applications, it is important to consider how to extract meaningful information from the collected data. For this reason, the third group of keywords focuses on concepts related to textual analysis, data mining and information extraction, essential elements for deriving value from data collected online. Finally, the fourth group of keywords includes terms related to various aspects of business administration, which can be influenced by the combined use of web scraping and AI (e.g. impacts on business operations, human resources management, marketing, sales).

The combination of these keywords was implemented using accurate mathematical logical connectors (Boolean and Proximity operators) such that the terms within each group are connected by the "OR" operator and each group of keywords is logically

Table 5 Area of analysis and inclusion criteria used to support results interpretation

Area of analysis	Inclusion criteria
Research trends over time	Temporal filters are applied (e.g. before and after 2017) to identify publications representing the different chronological orientations
Document type and Subject areas	Specific contributions that respond to these filters are selected, so as to offer concrete examples capable of enhancing and contextualizing the quantitative data
Geographical analysis	Documents are identified that directly dealt with issues related to distribution and geopolitical strategies associated with technological developments
Co-occurrences keyword analysis	Documents with explicit references to keywords relevant from a business perspective are considered, (e.g., sentiment analysis for sales optimization and ethical implications)

Table 6 Query structure used for Scopus research and properly adapted for WOS

TITLE-ABS-KEY
((("web") W/2 ("scrap*" OR "craw*" OR "harvest*" OR "data extract*" OR "data harvest*" OR "data scrap*" OR "automat* scrap*" OR "information extract*")) OR (“screen scrap*" OR "automat* web scrap*" OR "spider*"))
AND
("Machine learning" OR "ML" OR "artificial intelligence" OR "AI" OR "natural language process*" OR "NLP" OR "natural language generat*" OR "NLG" OR "artificial neural network" OR "ANN" OR "neural network" OR "deep neural network" OR "DNN" OR "recurrent neural network" OR "RNN" OR "unsupervised learn*" OR "convolutional neural network" OR "CNN" OR "supervised learn*" OR "learning algorithm" OR "convolutional neural network" OR "CNN" OR "SVM" OR "support vector machine")
AND
("text data" OR "Text mining" OR "data mining" OR "web mining" OR "web data mining" OR "knowledge discovery from text" OR "KDT" OR "information extraction" OR "IE" OR "information retrieval" OR "IR" OR "data enrich*" OR "data augment*" OR "data extract*" OR "data collect*" OR "semantic analysis" OR "sentiment analysis" OR "sentiment recognition" OR "text analysis" OR "data analysis" OR "information retrieval" OR "IR" OR "big data" OR "Text Summarization" OR "Text Generation" OR "Market Intelligence" OR "Predictive Analytics" OR "Reputation Monitoring")
AND
("Business" OR "firm" OR "enterprise" OR "company" OR "organization" OR "manufactur*" OR "industry" OR "market*" OR "account*" OR "sales" OR "decision making" OR "knowledge manag*" OR "strategic planning" OR "financial" OR "HRM" OR "human resource manag*" OR "SCM" OR "supply chain manag*" OR "risk management" OR "retail" OR "CRM" OR "customer relationship manag*" OR "insurance")

connected with the “AND” operator; the Proximity operator “W/2” is used in the first group of keywords in order to broaden the pool of results related to web scraping. By building the query with these operators, the results obtained will simultaneously satisfy all the criteria of each group, making the search more focused and targeted. In the case of this study, we are looking for academic articles that deal with both Web Scraping and ML and AI, Text and Data Mining, along with Business topics.

Papers containing these keywords in the title, abstract, and keywords were searched in the Scopus and WoS electronic scientific database in April 2025, chosen for their robustness as a bibliographic data source (Pranckutė, 2021). Subsequently, following the filtering process previously described in Table 4, the initial sample of 606 articles from Scopus and 217 papers from WOS was refined to 567 due to the document type filtering (i.e., notes, dataset and conference proceedings excluded), the restriction to English language and duplicate removal. All results were exported in CSV and RIS files, retaining all necessary information for conducting the subsequent analysis, including title, authors, year, country, document type, subject area, abstract, keywords and citation count.

4.1 Performance analysis

For the performance analysis of the data sample, several indicators were chosen to be analysed with the aim of answering the research questions. In particular, the number of publications and citations concerning the combined use of web scraping and AI was chosen to be analysed to obtain information on the interest and growth of research in this field. On the other hand, the temporal analysis of this metric can indicate trends over time, such as the increase or decrease of attention on the topic in the academic community. The geographic distribution of the publications provides insight into the geographical areas where there is greater interest or expertise in the combined use of web scraping and AI (Noyons et al. 1999). Furthermore, classifying papers according to type (e.g. articles, conference papers) provides insight into the most common sources of information in this field and can help identify the most authoritative and influential publications on the topic (Noyons et al. 1999). For instance, the prevalence of conference papers may signal a more immediate and dynamic exchange of ideas, while journal articles signify a deeper exploration and validation of concepts. Finally, by examining subject areas, it is possible to identify specific disciplines where these technologies are more widely studied, helping to contextualise the combined use of web scraping and AI in specific domains (Noyons et al. 1999).

4.2 Science mapping analysis

In the science mapping phase, a co-occurrence analysis of keywords in the title and abstract of the collected articles was conducted using the VOSviewer application, as a recognised tool that provides a low-dimensional visualisation to accurately reflect similarities between objects (Eck and Waltman 2007). Additionally, since it is not influenced by researcher subjectivity, it can help to decrease reviewer bias during research (van Oorschot et al. 2018). Specifically, this analysis achieves the following objectives: i) comprehend the most recurring topics in the fields of analysis; ii) identify relationships among them; iii) highlight emerging themes and trends over time.

Moreover, network analysis and graphical exploration of textual data were conducted using VOSviewer software, known for its ability to create easily interpretable co-occurrence maps on a large scale (van Oorschot et al. 2018). Following the guidelines of Donthu (2021a) and Fahimnia (2015), the analysis focused on counting recurring terms in titles, abstracts and keywords, with a minimum frequency of five. By carrying out a preliminary merge of the extracted keywords and creating a specific thesaurus, out of the 4001 terms examined, 51 meet the threshold, thus contributing to a clear and accurate delineation of the map of conceptual relationships within the extensive range of considered documents. The software offers three visualisations (i.e., network, overlay, and density visualisations) that enabled the identification of thematic clusters related to the combined use of web scraping and AI-based models in business and their evolution over time with the aim of answering the research questions of this study.

5 Results

5.1 Performance analysis

First, the quantity of documents published on Scopus and WoS over the years were explored (Fig. 1). This analysis was carried out with Excel and Python code using the Pandas, Matplotlib and NumPy libraries to generate graphs and identify relevant time spans. Specifically, we fit a linear regression model to the publication counts to identify the overall trend. In this way, a first significant period from 1998 to 2016 (T1) and a second significant period from 2017 to 2025 (T2) were identified.

Looking at the first time span T1, the graph analysis reveals that, from 1998 to 2008, the production of studies in the field of web scraping combined to AI-based models is negligible. However, from 2008 to 2016, a gradual increase occurs, indicating a growing interest in exploring this field. The Compound Annual Growth Rate (CAGR) calculated from 1998 to 2016 is 15.79%, indicating a steady growth rate in the number of documents published annually. This suggests a consistent increase in interest and research output over these years. During T1, the field was characterized by experimental exploratory engagements with web scraping technologies; therefore, the scant publication output, particularly before 2010, corresponds to a phase in which scraping was technically rudimentary and legally ambiguous (Craven et al. 2000; Navigli and Velardi 2005). This phase reflects a prototypical phase in which web scraping had not yet been systematized into a recognized research program and AI was still in its early stages of research (Cao 2009).

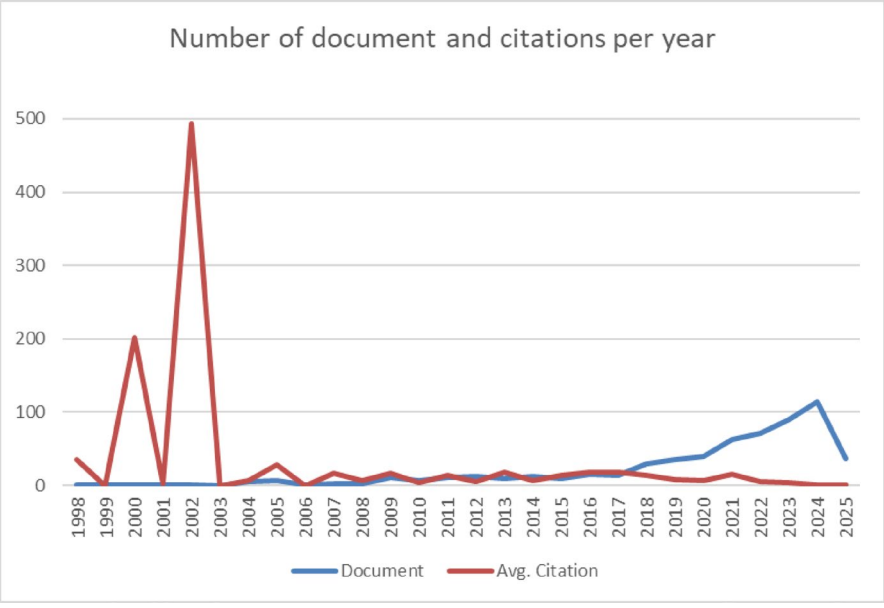


Fig. 1 Distribution of papers and number of citation by years

It is interesting to note that, the second time span T2 from 2017 to nowadays, there has been a rapid and steady growth in the number of studies related to this domain, with a CAGR of 35.78%.

T2 coincides with the integration of transformation architectures (e.g., BERT, GPT) (Althar and Samanta 2022) and the operationalization of scraping within larger, data-intensive pipelines (Moon et al. 2018; Mudzimu and Litvine 2019). The change in CAGR is symptomatic of a qualitative leap in conceptual legitimacy and scalability. In this period, scraping ceases to be a mere auxiliary technique and becomes part of a broader set of AI-based knowledge extraction methods. Furthermore, this phase is deeply intertwined with the spread of open-source AI models (e.g., Hugging Face transformers) and the growing pressure for online content extraction as a form of competitive tool in areas such as marketing. Therefore, the acceleration in publication volume reflects a convergence between technological development and the transfer of solutions to the market, such as the model for market trend assessment (Mudzimu and Litvine 2019) and the analysis of public opinions on products (Yi et al. 2023).

Citation dynamics further illustrate this maturation since older papers in T1 have higher citations due to their longer presence in the literature, allowing more time for other researchers to reference them. Conversely, the dip in citations for recent papers in T2 is expected since they have not had enough time to be widely cited. However, as these papers address more current and advanced applications of web scraping and AI, they will likely see an increase in citations over the next few years. More detail about number of publications and citations over years are available in Table 7.

The evolution in document type (Fig. 2) between T1 (1998–2016) and T2 (2017–2025) reflects a transformation in the scientific status and disciplinary anchoring of research on the integration of web scraping and AI.

In T1, the predominance of conference papers (66.4%) signals a nascent field, characterized by low entry thresholds, rapid feedback loops, and experimental openness (Klein et al. 2011; Malik and Rizvi 2011). Conferences served as incubators of methodological innovation, offering informal peer validation in the absence of codified standards. The relatively modest presence of journal articles (31.8%) suggests that, although some lines of research had reached a level of methodological consolidation (Fernández-Villamor et al. 2012; Petasis et al. 2014), the field had not yet achieved full epistemic legitimacy within academia.

In contrast, the publication landscape of T2 reveals an increase in journal articles (42.6%), signalling a point of transition from speculative experimentation to methodological stabilization. More comprehensive and refined studies are published (Reddy et al. 2021; Shaikh et al. 2021; Yi et al. 2023), indicating consolidated methodologies and findings. This shift signals a reconfiguration of scientific credibility regimes, where contributions must now align with standards of reproducibility, replicability, and conceptual rigor. The parallel emergence of book denotes both academic consolidation and curricular integration, as knowledge about web scraping and AI becomes a subject of teaching (Ball and Rague 2022).

From a disciplinary perspective, the distribution of the sample of articles by subject area in periods T1 and T2 clearly shows the evolution of the research focus and applications of web scraping and AI-based technologies (Fig. 3).

Table 7 Insight on number of publications and citations about web scraping and AI integration

Time Span	CAGR	Sub-period	Insight on Number of Publications	Insight on Number of Citation
T1	22.12%	Pre-2010	The number of documents per year is relatively low and sporadic, indicating early stages of interest and exploration in the integration of web scraping with AI models	Citations are scattered and relatively low, corresponding with the lower number of publications and their nascent stage
T1	9.05%	From 2010 to 2016	There is a gradual increase in the number of publications, suggesting growing recognition of the potential applications and benefits of web scraping and AI integration	Citations increase, but remain irregular, reflecting the gradual accumulation of references as the field gains traction
T2	35.78%	From 2017 to 2025	There is a significant and consistent rise in the number of publications, indicating a heightened interest and investment in research related to web scraping and AI	Citations peak around 2020, but there is a noticeable drop-off in the most recent years. This trend is typical as more recent publications have not yet had enough time to be widely cited

In T1, "Computer Science" (76%) dominates the scientific landscape. In this initial phase, most research is focused on machine learning applied to the construction of knowledge bases starting from the World Wide Web (Craven et al. 2000). Disciplines such as "Engineering" (17%) also show a growing interest in the application of these technologies to data collection and analysis, with studies focusing, for instance, on the identification of named entities in a university intranet (Althobaiti et al. 2012). The presence of "Decision Sciences" (16%) and "Mathematics" (13%) is linked to the development of advanced algorithmic approaches to NLP, such as the formalisation of structural semantic links for terminological disambiguation (Navigli and Velardi 2005), and their application in decision support systems that integrate expert

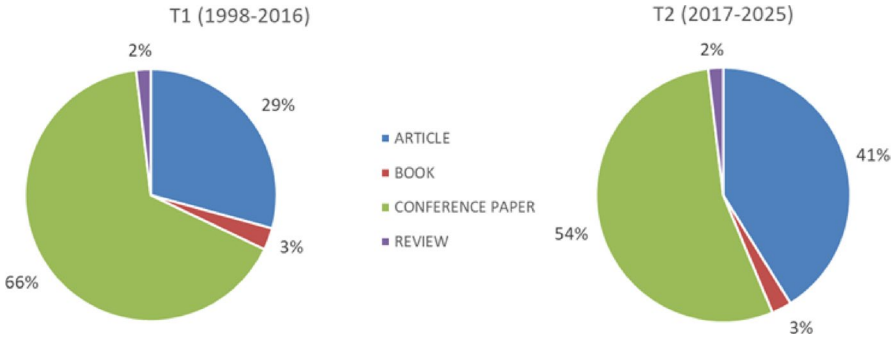


Fig. 2 Distribution of papers by document type over T1 and T2

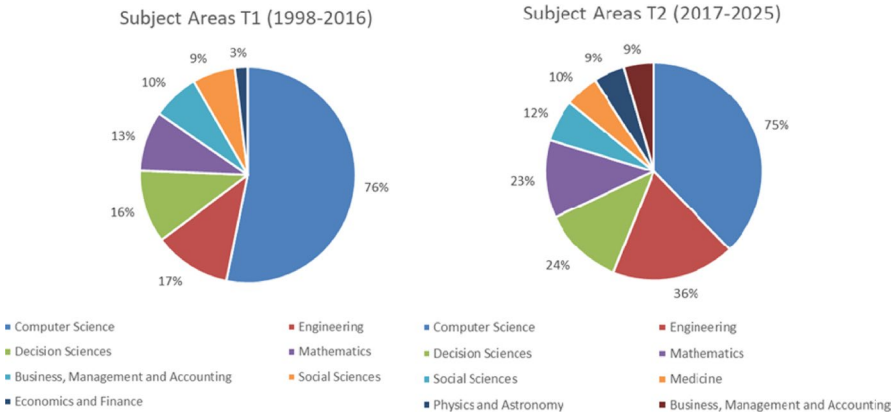


Fig. 3 Distribution of papers per subject area over T1 and T2

knowledge and data collected through multilingual web crawling in lead qualification processes (D'Haen et al. 2016).

In the medical field, studies are still limited and mainly concern pioneering work on filtering large amounts of social media data to build warning systems for adverse drug reactions (Yang et al. 2015). As far as the social science are concerned, there are only some marginal contributions related to socio-technological perspectives on the history of data (Robinson and Bawden 2017), while studies on web scraping and AI applied to environmental sciences are completely absent.

In T2, a significant broadening of the disciplines involved can be observed. "Computer science" (75%) continues to play a dominant role, as expected given the digital nature of the technologies concerned, but the focus of research is diversifying, including applications such as the classification of food products (Hillen 2019) or prediction on future skills for labour placement (Mathew et al. 2018).

In parallel, the relevance of "Engineering" (36%) and "Mathematics" (23%) is increasing, focusing on technical aspects such as the development of methodological frameworks for AI-based text analysis in digital media (Cordeiro et al. 2025) and the intelligence management for cybersecurity (Spyros et al. 2025). Research in decision

sciences (24%) is increasingly focused on the automated analysis of opinions and sentiments, such as studies on the impact of news on stock prices (Gupta et al. 2025).

In "Medicine" (10%), applications are becoming more consistent, including the development of computer-aided diagnostic systems for breast cancer detection (Dihmani et al. 2024) and the use of AI to combat health misinformation (Shinde et al. 2022). At the same time, the social sciences (10%) are beginning to explore innovative topics such as the ethics of consumption influenced by artificial intelligence embodied by virtual influencers (Marti-Ochoa et al. 2025).

Finally, but still to a lesser extent, "Environmental Sciences" (3%) appear with studies using AI models to monitor activities related to the circular economy and employment (Borms et al. 2024). However, there is a lack of research analysing the direct environmental impact of the adoption of technologies, such as web scraping and AI, highlighting another knowledge gap that needs to be filled.

Finally, the production of publications over the years by country (Fig. 4) reveals more than a simple disparity in the volume of research.

The concentration of production in China, India, and the United States reflects economic investment in research and development and signals the consolidation of techno-cognitive sovereignty. In these contexts, AI and web scraping are integrated components of broader national strategies that aim to combine data extraction with algorithmic governance (Eisenberg and Abhari 2023). This raises concerns about who owns the means of automated knowledge extraction, who defines the policies and ethics around data extraction and its use.

For instance, China stands out for its centralized strategy, supported by strong public policies and systemic adoption of AI in public and private sectors (Allen 2019). This produces a large amount of publications, but also a very control-oriented and efficient vision of information systems.

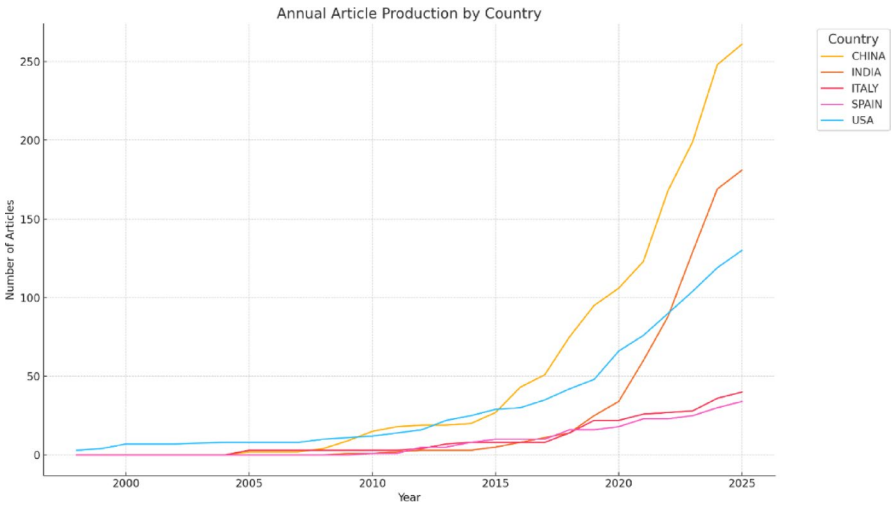


Fig. 4 Annual article production by Country

India shows a different trajectory as the number of publications reflects the country's enormous technical capacity, combined with a strong engineering culture and a large availability of trained human capital (Sahoo and Urkude 2025).

The United States remains central both for the amount of research and for the role it plays in defining global standards and orientations. Much of the software infrastructure used in the AI field (e.g., frameworks, datasets, APIs) is developed or supported by US companies and institutions and therefore has a strong influence on policies and data processing for other countries as well (U.S. House of Representatives 2020).

In Europe, the situation is more fragmented as countries such as Italy, Spain, Germany, and France contribute significantly but with a greater focus on ethical, regulatory and applicative aspects. For instance, much European research focuses on responsible uses of AI (Baldassarre et al. 2024), interoperability of systems (Perisic et al. 2023), and privacy management in data collection and analysis processes (Furizal et al. 2024).

Moreover, data suggests a correlation between the number of publications and the level of economic and technological development in a country. Countries with advanced technological infrastructure and strong economies tend to have more publications, reflecting their capacity to support extensive research activities. Companies in leading countries are at the forefront of utilizing these technologies for competitive advantage, driving innovation in various sectors, such as marketing, finance, healthcare, and more.

However, the comparative analysis between the number of publications and the average number of citations per country (Table 8) reveals that the impact of research is not proportional to the quantity of articles produced. Countries such as China and India, which dominate in terms of volume (261 and 181 publications respectively), show rather low citation averages (9.4 for China and 3.7 for India). At the other extreme, we find cases such as Brazil (164.7 average citations per article but only 7 publications) and Bahrain (144 average citations and 2 publications), which obtain a very high citation impact. In high-production countries (China, India, USA), it is likely that we are witnessing a form of editorial inflation in which publications on topics already widely covered lead to a lower average impact. Furthermore, it is likely that even if certain countries contribute a lot of research, it may be the result of highly specific and sectoral studies with non-replicable results and therefore unlikely to garner a large number of citations.

Table 8 Total and average citation by Country

Country	Total Citation	Average paper citations
China	1069	9.4
Usa	660	16.5
Brazil	494	164.7
Italy	466	31.1
India	338	3.7
United Kingdom	154	15.4
Bahrain	144	144
Korea	119	13.2
Germany	95	9.5
Romania	60	10

5.2 Science mapping analysis

5.2.1 Network visualization

The network visualization (Fig. 5) of the co-occurrences analysis, shows three clusters, identified by different colours with recurring terms represented as nodes and connections as arches. The size of the node and the edge reflects the term's occurrence and connection strength, while centrality underscores its relevance to the cluster and the entire network.

The central positioning of key nodes such as ML, data mining, web scraping, NLP and AI highlights their fundamental role in the field of web scraping and AI integration, as expected.

Cluster 1 (red), comprising 24 terms, refers to "Intelligent Web Crawling". Major keywords are "data mining" (160), "natural language processing" (156), "web crawler" (138), "artificial intelligence" (87), "information retrieval" (84), "classification" (77), "websites" (98). This is where the integration between web scraping and AI takes shape, understood as an automatic pipeline for the extraction, classification and organization of digital information. It is significant that web crawler has a weight comparable to NLP and data mining because this indicates that the heart of research is still anchored to the web as a direct source of textual data, and not just as a passive repository. Furthermore, data mining, text mining, text analysis, semantic web technology and NLP describe the transformation of these raw data into treatable representations such as structured classifications (Reddy et al. 2021).

The cluster also includes terms such as decision making, decision support systems, intelligent agents, knowledge-based systems and machine learning techniques, indi-

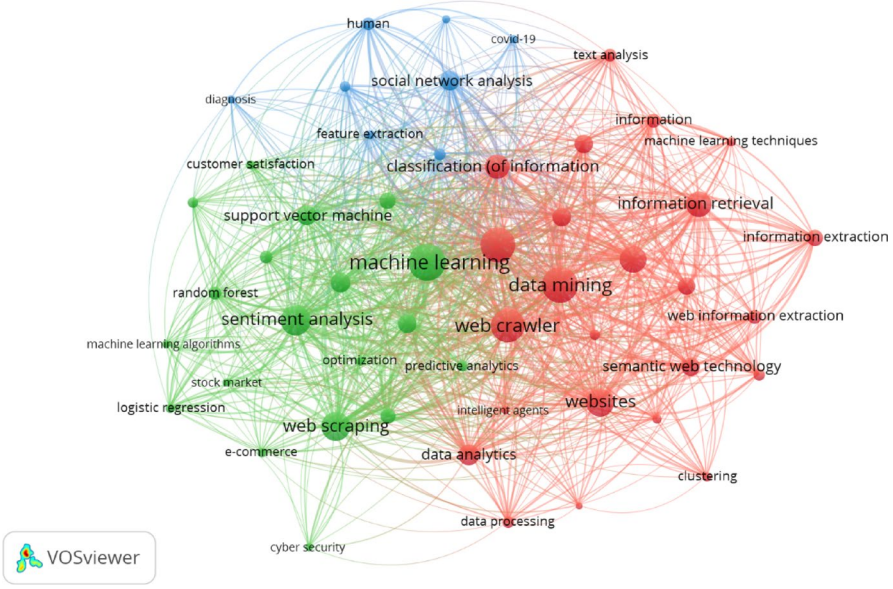


Fig. 5 Network visualization

cating that the collected data are used to support decisions, automate responses, and build intelligent agents capable of acting autonomously. In particular, web crawling and NLP models are widely used to gather contextualized data on market insights useful to represent customer opinion (Kincl et al. 2013), to find targeted sales strategies (Chong et al. 2016), and to predict trends for informed investment decisions (Tong and Tian 2023). To support these considerations, words such as information extraction, clustering, and information management indicates an attempt to formalize the extracted knowledge, to make it interoperable and navigable.

However, data mining techniques need to become more scalable and adaptive as the volume and diversity of web data continue to grow exponentially (Tanasescu et al. 2022). Moreover, considering the rapid evolution of language and slang, learning models need to be adaptable to new language patterns in real time without requiring extensive retraining (Corallo et al. 2020). Another critical area is the integration of real-time data processing capabilities to enhance the timeliness and relevance of the extracted insights, otherwise classifications may become obsolete and distorted (Bricongne et al. 2023).

Cluster 2 (green), contains of 19 terms, addresses the "Advanced Sentiment Analysis". The main keywords include machine learning (170 occurrences), sentiment analysis (118), web scraping (105), deep learning (51), support vector machine (50), neural network (33), convolutional neural network (21), random forest (20), logistic regression (13) and large language model (14). Unlike the first cluster that focuses on data collection and structuring, this cluster focuses on the different architectures of AI models for trend forecasting analysis, opinions and anomaly detection in information systems.

The inclusion of different neural networks architecture, including LLM, also suggests the connection with sentiment analysis thanks to the transformative architectures underlying models such as GPT (OpenAI et al. 2023) that amplify the capacity of semantic analysis and sentiment recognition from online data (Zhang et al. 2023). In fact, the cluster also features several market-related terms (e.g. commerce, e-commerce, finance, stock market, customer satisfaction, predictive analytics, and optimization) that denote a business-driven orientation of these algorithms. This result suggests that the technologies in question are functional to performance, forecasting, and automation of business decision-making processes.

Sentiment analysis plays a crucial role in this cluster since it links AI technologies to the attempt to interpret, classify, and measure emotions and opinions expressed online, often on social platforms, forums, or reviews. This theme is central both to e-commerce and to applications in the political, financial, and corporate reputation management fields. In this cluster, a significant challenge emerges regarding the accuracy and reliability of sentiment analysis models, particularly when dealing with the nuances of human language, such as sarcasm, idioms and cultural differences (Azad et al. 2018).

The co-occurrence with web scraping clearly shows that the primary source of these analyses is the open web that contains spontaneous opinions collected from comments, reviews, tweets, and posts. Moreover, the accuracy and reliability of AI models, such as neural networks used in data analysis, must be improved to minimize errors and biases in interpreting complex data (Choi et al. 2019).

It is therefore an automation of social listening, which can produce advantages, but also significant ethical problems related to privacy, implicit consent, and the semantic quality of the data collected (Srinath et al. 2021). Of notable importance is the presence of the term cyber security (8 occurrences) which reminds us that digital environments involve vulnerabilities and if we deal with data collection and analysis, it is inevitable to take into account both the regulations for the use of information and that it is necessary to protect it by safeguarding consumers (Furizal et al. 2024).

Cluster 3 (blue), refers to "Digital Social Network" as it contains 8 highly significant keywords, i.e.: social network analysis (50 occurrences), human (23), algorithm (17), classification models (16), feature extraction (13), covid-19 (10), diagnosis (8), and information dissemination (9).

The strongest keyword in the cluster is social network analysis, which serves as a linchpin to interpret the role of digital networks in the spread of information and disinformation, especially in critical scenarios such as the pandemic (Thota and Ramez 2021). These are studies that have attempted to monitor the spread of online content related to public health (Anto-Ocrah et al. 2025), analyzing both the circulation of correct information and the phenomenon of misinformation (Shinde et al. 2022).

The presence of the keywords classification models, algorithm, and diagnosis indicate an extensive use of computational models to identify significant patterns in the data. The inclusion of information dissemination reinforces this reading by underlining a growing interest not only in information itself, but also in the dynamics of diffusion and the impact it has on human behavior (Kafeza et al. 2023).

In particular, the presence of the keyword human is a reminder that algorithms produce classifications that directly affect people. The relationship between algorithms and humans in this cluster is explored in contexts where the classification of information has direct effects on people's lives, such as in health diagnosis (Abdullahi et al. 2024) or in managing communication in emergency situations (Kumar et al. 2021). This is not a neutral act, it is a decision that can have practical, social or even health consequences.

What is striking, looking across the three clusters, is that the entire research ecosystem appears to be strongly oriented towards operational efficiency and the economic valorization of AI. The implicit objective seems to be to extract predictive value from large volumes of data to sell more products (Marti-Ochoa et al. 2025), optimize processes (Abhijith et al. 2024), and improve business performance (Vording 2021). This non-orientation is in itself negative because it promotes economic growth, creates job opportunities, and supports innovation. However, the same automation that enhances business can in fact replace human tasks, leading to an accelerated redefinition of the labor market (Navas-Loro et al. 2023). The skills required are rapidly changing, and with them the need for continuous training, which entails economic and social costs, but also cultural resistance from those who are unable or unwilling to adapt to these rhythms (Korzynski et al. 2023). Faced with these transformations, the silence of the literature on the societal impact of AI combined to web scraping is surprising because there seems to be no reflection on how these technologies reconfigure the relationship between man, machine, and knowledge. Equally absent is the explicit reference to legal and regulatory issues, starting with those related to data collection and processing. One would have expected to find a strong focus on

informed consent, especially considering that the data collected comes from posts, blogs, conversations, multimedia content and online recordings. In most cases, users are not really aware that the information they share becomes raw material for predictive algorithms (Hari Krishna et al. 2021; Marti-Ochoa et al. 2025). This problem is amplified in the most delicate contexts, such as health data or critical infrastructures, where one would expect much more explicit attention to data protection and process transparency (Bala et al. 2024).

Finally, although the term cybersecurity emerges, a reflection on the malicious use of the extracted information remains completely marginal. The enhancement of NLP capabilities makes phishing and social engineering attacks increasingly credible and customizable, with concrete consequences for both individual security and corporate reputation and vulnerability (Campbell et al. 2022; Hegde et al. 2023). In this sense, the literature seems to ignore that the same technology that today is used to optimize marketing can be used to orchestrate scams with extreme realism.

The main keywords of each cluster are reported in Table 9 with the interpretation of possible applications and challenges.

5.2.2 Overlay visualization

The overlay map presented in Fig. 6 offers a temporal perspective, enhancing the understanding of term co-occurrence dynamics over time. The map distinguishes the terms chronological dimension by using distinct colours, blue for older terms, green for intermediate terms, and yellow for recent terms. The time interval selected, spanning from 2014 to 2025, enables a balanced observation of term variations throughout the specified period.

The evolution from foundational data management (2014-2017) to sophisticated data analysis and AI applications (2017-2019) and finally to advanced techniques and real-world applications (2019-present) highlights the growing synergy between web scraping and AI in extracting valuable insights from vast amounts of web data. This reflects the continuous development in leveraging technological advancements to address emerging challenges and opportunities in the digital landscape.

During the period between 2014 and 2017, a clear dominance of concepts such as information retrieval and extraction, semantic web, knowledge management, and ML emerged. This early phase indicates an emphasis on organizing and retrieving data from the web. In particular, the research is focused on building robust frameworks for managing and extracting structured information (Fernández-Villamor et al. 2012; Sleiman and Corchuelo 2014), which is essential for any subsequent data analysis. The dominance of these concepts suggests that the initial challenge was to develop systems capable of efficiently handling the vast and growing amounts of web data. This period laid the groundwork for more advanced applications by establishing the necessary infrastructure for data management and retrieval, making it possible to harness the full potential of web data for various purposes. However, these advancement faced significant challenges related to the scalability and dynamic nature of web data as early models struggled with handling the vast volumes and varied formats of web data (Dhillon et al. 2011), leaving many potential insights unexplored.

Table 9 Key applications and challenges related to identified clusters

Cluster	Main keywords	Key Applications	Main Challenges
Intelligent Web Crawling	Data mining NLP Web crawler AI Information retrieval Websites	Transforming raw web data into structured formats, enhancing data analysis capabilities with AI, managing large data volumes efficiently	- Integrating AI with traditional systems in a interoperable way - Scalability of data mining techniques and real time processing - Ensuring robust data security and privacy protection
Advanced Sentiment Analysis	ML Sentiment analysis Web scraping DL SVM CNN	Understanding customer opinions and behaviours, optimizing sales strategies, predicting market trends. Applications in commerce, sales, forecasting, and financial markets	- Handling diverse and unstructured and heterogeneous data- Accurately interpreting human language nuances (e.g., sarcasm, cultural differences)
Digital Social Network	Social network analysis Human Algorithm Covid-19 Information dissemination	Analysing human behaviour in digital contexts, especially during crisis situations (e.g., pandemics), to monitor how information circulates and to support the management of public communication	- Ethical and privacy issues related to data collection and decision-making usage- Ensuring robust data security and privacy protection

Since 2017, the research lexicon has been enriched with words such as data mining, natural language processing, web crawler and text mining, which signal the adoption of more advanced models for automatic text analysis in line with the introduction of transformative neural networks and the self-attention mechanism (Vaswani et al. 2017). In this period there is the integration of scraping with classifiers and ML algorithm. The centrality assumed by decision making and predictive analytics shows that the focus is rapidly shifting from data processing to the operational valorisation of the information collected, with applications in decision support and forecasting.

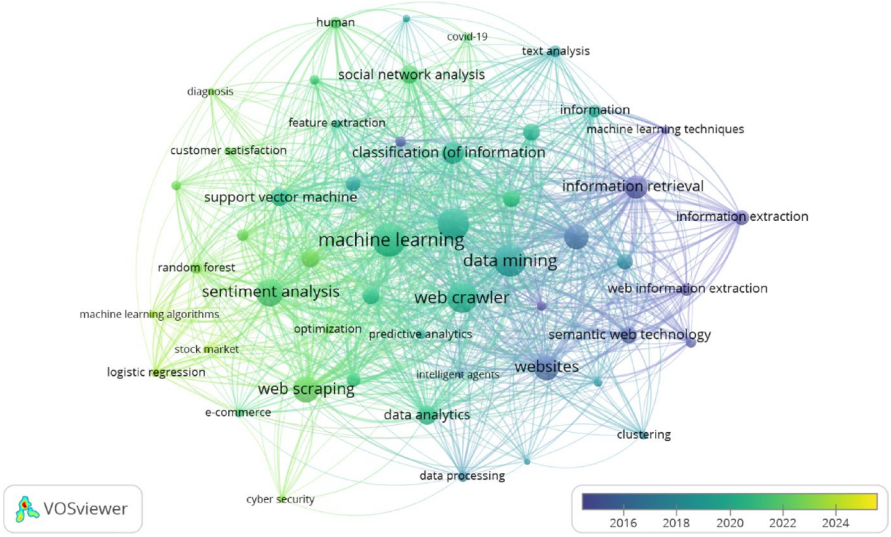


Fig. 6 Overlay visualization

In the middle period between 2017 and 2020, there has been a clear shift of interest toward research in NLP, learning systems, web crawling, and data mining marking the transition from foundational data management to more complex data analysis and understanding. This shift reflects a growing recognition of the importance of unstructured online information and the need to develop more sophisticated systems to manage and analyse these data (Baviskar et al. 2021). The focus on NLP indicates efforts to understand and process human language, a critical aspect of extracting meaningful insights from text data on the web. Moreover, learning systems and data mining highlight the implementation of advanced algorithms to discover patterns and knowledge from large datasets, while web crawling became a critical technology for automating the collection of vast amounts of web data, essential for feeding these advanced analytical systems. The NLP advancements improved the ability to process and understand human language, though challenges related to linguistic nuances and context remained (Chaudhary and Kakkar 2021). Furthermore, the implementation of web crawling technique allowed for automated data collection on a larger scale, but this also brought about ethical and legal concerns regarding data privacy and the potential misuse of collected information.

In more recent years - from 2020 to 2025 - concepts such as web crawling, sentiment analysis, text mining, neural networks, and DL have emerged underscoring the continuous development and adaptation of technologies to meet the ever-changing needs of research and practice in the digital era. Therefore, this recent focus indicates a growing interest in applying advanced data analysis techniques and sophisticated technologies to various domains, particularly those related to online social dynamics. Specifically, sentiment analysis and text mining demonstrate efforts to extract nuanced insights from unstructured text data, such as understanding public opinion and trends (Yi et al. 2023). Moreover, the use of neural networks and DL signifies the

adoption of cutting-edge AI technologies to enhance the accuracy and efficiency of data analysis (Mackey et al. 2020; Yi et al. 2023).

However, what is most striking is not only what has emerged, but above all what systematically remains on the margins. In a context in which AI is used to infer emotions, intentions and behaviours on a global scale, it is surprising to note the structural absence of reflections on the political limits of these practices. It is implicitly assumed that more data and more powerful models automatically produce better knowledge, but the fact that these technologies only describe the reality that can be measured and classified is overlooked. Furthermore, the total absence of keywords related to ethics, environmental sustainability, transparency, bias or accountability reveals a conceptual void that cannot be ignored. It seems that an advanced technical infrastructure is being built to read the world, without asking who it serves, who it excludes, and at what ecological and social cost. In this sense, the real risk is not so much the incorrect use of AI, but the "normal" but uncritical use of models that operate in the absence of thought about their transformative scope.

To highlight the thematic shifts, research implications, and conceptual gaps across the identified time period, Table 10 is presented as follows.

5.2.3 Density visualization

The density visualization depicted in Fig. 7 offers a compelling insight into the distribution and significance of terms across a vast array of analysed articles. Each keyword is distinguished by its unique label, with the surrounding area color-coded to indicate the density of articles in that specific position.

Notably, pivotal terms such as web crawling, data mining, ML, NLP, sentiment analysis, big data, and learning systems stand out as focal points of dense concentration, characterized by deep blue. These topics hold central positions within the extensive literature and scholarly debate surrounding web scraping and AI-based models, reflecting their critical importance and extensive exploration within the literature. The prominence of these terms underscores the significant focus on developing and refining technologies that automate and enhance data extraction, analysis, and interpretation processes. Moreover, the centrality of these concepts indicates a robust and ongoing interest in leveraging AI and ML to handle vast amounts of web data, extract meaningful patterns, and apply these insights across various domains.

In contrast, terms such as DL, forecasting, SVMs, text processing, sales, commerce, and finance exhibit a more dispersed distribution, with lighter shades of yellow indicating a lower concentration in specific research areas. This dispersion may indicate that while these topics are integral to the broader field, they do not command the same level of concentrated scholarly attention as the central themes. Their broader relevance across multiple disciplines implies that these concepts are applied in a wide range of applications, from specific technological advancements to practical implementations in business and finance.

While certain terms dominate the centre of the research landscape, this centrality may also reflect a form of conceptual inertia, where academic attention continually reinforces what is already widely explored, potentially at the expense of more disruptive or interdisciplinary directions. The peripheral positioning of commerce, finance,

Table 10 Web scraping and AI research trends evolution

Period	Key Concepts	Technological Focus	Implications	Underlying Gaps	References
2014–2017	Information retrieval, Semantic web, Knowledge management, ML	Building frameworks for structured data extraction and management	Enabled future AI applications by prioritizing data structuring capabilities	Scalability limitations and lack of adaptability to dynamic web data	Dhillon et al. (2011), Fernández-Villamor et al. (2012), Sleiman and Corchuelo (2014)
2017–2020	Data mining, NLP, Learning systems	Integration of scraping with ML, transition to unstructured data analysis, automated data collection	Shifted research from infrastructure to operational decision support and automation	Emerging ethical concerns around automated data collection not fully addressed	Baviskar et al. (2021), Chaudhary and Kakkar (2021), Vaswani et al. (2017)
2020–2025	Web scraping, Sentiment analysis, Text mining, Neural networks, DL	Advanced AI applications for text and sentiment analysis, focus on online social dynamics	Indicates a growing ambition to model social dynamics, but risks oversimplification	Systematic absence of discourse on ethics, sustainability, and accountability	Deepa et al. (2024), Mackey et al. (2020), Yi et al. (2023)

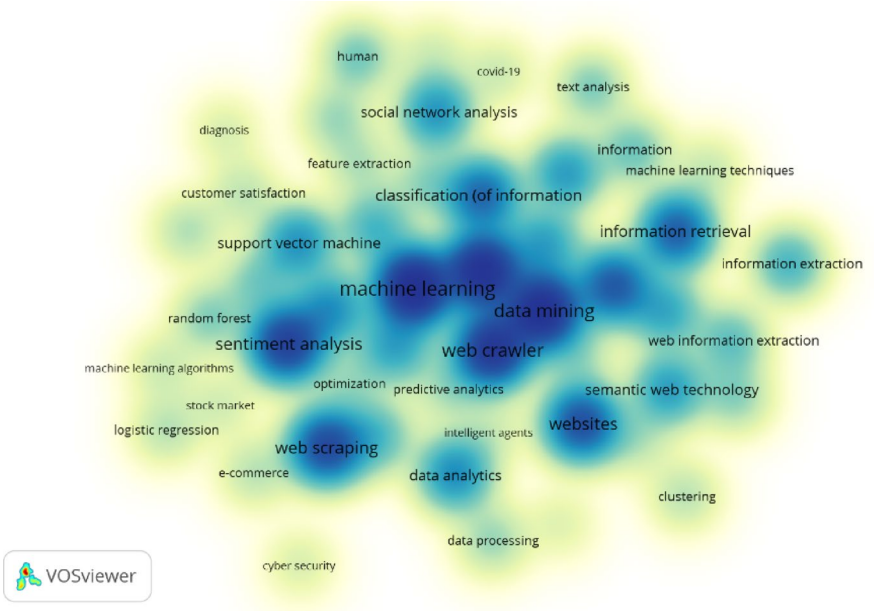


Fig. 7 Item density visualization

or forecasting may signal a hesitation to fully address the systemic consequences of decision-making automation in domains where errors incur tangible costs. For example, a sentiment analysis system deployed in the financial sector may misinterpret an online emotional fluctuation as a signal of market panic (Bhat and Jain 2024). Even in online commerce, faulty customer satisfaction analyses may lead to counterproductive decisions, such as discontinuing popular products among niche consumers that cannot be captured through generic models (Arsad et al. 2021; Dwivedi et al. 2023). Furthermore, the fragmented treatment of critical tools such as DL and SVM may indicate that, despite their technical power, they are treated as modular solutions rather than as objects of in-depth methodological scrutiny (Jiang 2022; Uma et al. 2022). The risk is that the uncritical adoption of these models leads to formally accurate but fragile results, that is, lacking a real understanding of why they work or fail in certain contexts. A deeper methodological reflection could instead contribute to developing more transparent, contextual and responsible practices of use. This requires a more reflexive attitude on what types of knowledge are prioritized, who benefits from them and what questions remain unanswered in the current research landscape.

6 Discussion

6.1 Research evolution and business implications

The combined use of web scraping and AI-based models has significantly evolved over the years, offering profound insights into business implications summarized and clustered in in Table 11. This evolution is reflected in the increasing number of publications and the diversity of research topics, as evidenced by the performance and science mapping analyses conducted in this study.

Initially, the focus was on establishing foundational knowledge and methodologies, as evidenced by the high proportion of conference papers (Fig. 2) and dominance of computer science (Fig. 3). Over time, the field has expanded and matured, with an increasing number of journal articles and a more diverse range of subject areas indicating the broadening applications and interdisciplinary nature of AI into different fields (e.g. Medicine, Social Science). These areas reflect the key role of the development of advanced tools, such as sophisticated decision support systems (Tong and Tian 2023) and data management systems (Hamada and Naizabayeva 2020), which enable reliable web scraping performance. We can identify two well-defined

Table 11 Evolution of web scraping and AI models over the time and relative business implications

Time Period	Keywords	Business Implications	Insights description
Pre-2017	- World Wide Web - Information extraction/ retrieval - Web mining - Search engines - AI	- e-commerce and real-time communication- Customer insights- Data collection challenges	- Opening to new competitive global markets- Analysing user-generated data to understand sentiments, opinions, tastes, preferences, and habits for targeted marketing- AI technologies emerged to enhance data collection from unstructured formats like text, audio, images, or video
Breakpoint 2017–2018	- NLP - Data/Text mining - Web scraping - ML - Information classification	- Improved data management and analysis- Real-time insights	- Handling unstructured web data more effectively with advanced algorithms and ML models- Rapid growth in online data drove demand for real-time insights, aiding business intelligence, decision-making, and strategic planning
Post-2017	- DL - CNNs - Sentiment analysis - SVMs - Sales - Forecasting - Commerce - Financial markets - Social media	- Enhanced NLP capabilities - Advanced data analysis - Consumer behaviour and market trends - Corporate reputation and workforce planning- Product strategy and customer experience- Operational efficiency and process automation- Innovative applications	- Improved sentiment analysis for understanding preferences, emotions, and opinions- Application of DL for sentiment analysis and social media mining- Analysing purchasing patterns and product perceptionOpinion analysis about companies and employee reviews- Reducing bias in product reviews, enhancing recommendation systems, and analysing market demand- Detecting incorrect sales data, classifying customer reviews, efficient online information collection and understanding knowledge exchange- Creating video recommender systems, automatic transcription via speech recognition, and developing virtual assistants for e-commerce

time periods in the evolution of web scraping and AI-based models: pre-2017 and post-2017.

In the pre-2017 period, research was concentrated on concepts related to the World Wide Web, the internet, information extraction, web mining, ontologies, search engines, information retrieval systems, and AI. Its was a time when the internet was becoming integral to everyday life, and companies were beginning to recognize its potential. The advent of e-commerce and real-time communication without geographical constraints opened new, competitive global markets (Rejeb et al. 2020). Concurrently, the rise of blogs, social media, online transactions, and digital communications generated vast amounts of data daily (Hogenboom et al. 2012). This data, rich with information, became invaluable to the global marketplace, further intensifying competition. Companies saw the potential to harness and analyse this data to gain insights of user-generated data reflecting sentiments, opinions, tastes, preferences, and habits that could guide business strategies such as targeted marketing and better customer service (Ceri et al. 2013) and improvement of customer experiences (Xu et al. 2011). This agility is crucial in today's fast-paced business environment, helping organizations predict trends, identify risks, and seize opportunities. Companies thus had every incentive to fund research into data collection and analysis, which is reflected in the geographical distribution of research papers in this field. However, this data often exists in unstructured formats like text, audio, images, or video, making it challenging for web scraping technologies to extract useful information from complex web structures (Navigli and Velardi 2005). Here, AI models emerged to enhance the data collection process by training models to accurately identify and extract desired information (Adda 2011).

The 2017–2018 period marked a breakpoint with the emergence of new concepts such as NLP, data mining, web scraping, ML, and information classification (Fig. 6). This shift was driven by the need to manage and analyse unstructured web data more effectively. The rapid growth of online data and the demand for real-time insights necessitated the development of more advanced AI models. Technological advancements, particularly in neural networks (Sleiman and Corchuelo 2014) and the introduction of the "Self-Attention" mechanism (Vaswani et al. 2017), propelled this shift. This period of dynamic growth in research highlighted the increasing interest in leveraging these technologies for business intelligence, decision-making, and strategic planning (Baviskar et al. 2021).

In the post-2017 period, research continued to evolve with the emergence of DL, CNNs, sentiment analysis, SVMs, and continued interest in web scraping. Additionally, business-related terms such as sales, forecasting, commerce, financial markets, and social media became prominent. This shift indicates that research is moving towards practical applications for companies such as document management (Moon et al. 2018), sustainable tourism (Blasi et al. 2024), urban pollution (Tasnim et al. 2023) and pharmaceutical therapy perception (Schück et al. 2021).

The advancements in neural networks architectures have significantly improved NLP models capabilities, facilitating sentiment analysis through social media mining, which is crucial for understanding preferences, emotions, and opinions (Abhijith et al. 2024). Sentiment analysis enables companies to analyse purchasing patterns to improve marketing strategies (Palencia-Olivar 2022), detect consumer perception

of producs (Arsad et al. 2021) and predict the penetration of new products on the market (Afandizadeh et al. 2023). Additionally, companies can use these technologies to analyse public opinion about company products (Yi et al. 2023) and employee reviews (Adib et al. 2023) providing insights into corporate reputation and workplace dynamics, while identifying emerging job competencies trends that can help in strategic workforce planning and development (Sheriff and Sevukan 2023). On the other hand, reducing bias in product reviews (Suganya and Vijayarani 2020) and enhancing recommendation systems (Venkata et al. 2022) improve customer experiences, and analysing market demand (Hossain et al. 2022) informs product strategy. Detecting incorrect sales data (Mackey et al. 2020) and classifying customer reviews (Haris et al. 2023) enhance data accuracy and reliability; while, efficiently collecting online information (Fabi et al. 2022), understanding knowledge exchange (Zhao and Yang 2023), and improving web data extraction processes (Patnaik and Babu 2022) streamline operations. Additionally, business applications include creating video recommender systems (Boppana and Sandhya 2023), automatic transcription via speech recognition (Apturkar et al. 2020), and developing virtual assistants for e-commerce (Prajwal et al. 2019), further showcasing the versatile and transformative impact of these technologies.

6.1.1 Systemic implications of data-driven practices

Although the proliferation of business-oriented implications clearly demonstrates the practical relevance of combining web scraping and AI, this dominant orientation also reveals a deeper imbalance. Most research implicitly assumes that more data, faster automation (Ghadekar et al. 2023; Tasnim et al. 2023) and better prediction (Bhat and Jain 2024; Dawson et al. 2020) are inherently positive goals, but little attention is paid to the socio-economic asymmetries that these technologies might reinforce (Cordeiro et al. 2025; Furizal et al. 2024). As AI models shape the way companies interpret behavior, take consumer preferences (Abhijith et al. 2024) and make strategic or hiring decisions (Ghadekar et al. 2023), they risk encoding systemic biases, narrowing options and optimizing only for those already visible to the system (Sharma et al. 2020). Moreover, the same tools that improve customer targeting may accelerate forms of algorithmic invisibility for individuals who do not fit pre-existing behavioral patterns, rendering them effectively irrelevant to certain market decisions (Dwivedi et al. 2023).

This overlooks the displacement effects on labor (Navas-Loro et al. 2023), the increasing dependence on highly specialized technical skills and the resulting pressure on education systems and workers to constantly adapt (Deepa et al. 2024). In this context, rapid technological change is not only an economic challenge, but also a cultural and psychological one, raising questions about the pace of change that societies can sustainably absorb and the psychological burden of continuous professional updating (Deepa et al. 2024; Yu et al. 2023).

Equally absent from this business-focused vision is a substantive confrontation with legal and ethical frameworks. The widespread use of web scraping assumes the web as a freely exploitable data resource, but ignores the legal ambiguity of large-scale data mining, especially with regard to informed consent (Porcelli et al. 2024). Most

users are unaware that their content (posts, reviews, interactions) is being collected and analyzed to train models or generate business insights (Pantano et al. 2019). The absence of clear regulatory signals or ethical safeguards, particularly when handling sensitive data (e.g. health, work, critical infrastructure), exposes both companies and individuals to risks that current research barely addresses (Knieps 2024).

Despite the recurrent use of AI in analyzing opinions and extracting reviews, there is little discussion of the misuse of these same capabilities. Advances in NLP make it increasingly easy to generate realistic fake reviews, simulate public opinion or automate phishing campaigns with unprecedented linguistic precision, posing a threat not only to consumers, but also to corporate reputation and public trust (Gupta et al. 2023).

Even if the scraped data is used and protected according to the rules, there is no evidence of vulnerability to inference attacks, where a malicious actor can reconstruct the original data or deliberately poison it to manipulate the results of the algorithms (Furizal et al. 2024). Furthermore, this research has focused more on text analysis, but unstructured data also includes multimedia files such as audio, images and videos, which makes deep faking attacks, defined as the creation of realistic multimedia content using characteristics of existing people that are used for criminal intent or to ruin reputation (Campbell et al. 2022), even more important and concrete, making it even more important to develop research on scraping and AI in a more responsible and risk-aware way.

The combination of scraping, social network analysis and AI models suggests a new form of knowledge management based on digital technologies used for rapid, automated or semi-automated decision-making (Tong and Tian 2023). However, if choices are the result of unsupervised processes, or of AI models that learn from noisy data, it becomes difficult to understand the rational of the output leading to a progressive delegation of cognitive authority to machines, without adequate human control or transparency in processes (Baldassarre et al. 2024; Kumar et al. 2021).

To highlight how technical practices of process automation and optimization intersect with legal, ethical and social risks, Table 12 summarizes the main systemic implications for companies arising from the combined use of web scraping and AI models.

6.2 Research gaps and future research agenda

In light of the results of the bibliometric analysis, it was possible to outline a future research agenda (Table 13) aimed at filling the main gaps identified in the literature.

6.2.1 Technical insights

One of the most critical and widely discussed nodes in the integration of web scraping and AI concerns data quality and model transparency. Web scraping systems do not discriminate between reliable sources and inaccurate or manipulated content incorporating also errors, misinformation and cultural bias (Kumar et al. 2021). When these data feed into AI models, the risk is to amplify existing biases, consolidating distorted or even socially discriminatory narratives (Dwivedi et al. 2023). The ‘black box’

Table 12 Systemic implications for companies of the combined use of web scraping and AI models

Domain	Implications for Companies	References
Bias and Visibility	Risk of reinforcing systemic biases and exclusion of marginal profiles	Sharma et al. (2020)
Labor and Skills Pressure	Demand increase for technical expertise, constant re-skilling and parallel displacement of jobs	Deepa et al. (2024)
Legal Ambiguity	Web scraping often occurs in a grey legal zone with unclear consent mechanisms	Porcelli et al. (2024)
Regulatory Gaps	Lack of safeguards for sensitive data and transparency in AI model outputs	Furizal et al. (2024)
Security and Data Integrity	Exposure to cyber risks and weak protections against data manipulation	Knieps (2024)
Reputation and Trust Risk	Generation of fake reviews, opinion simulation, and linguistic phishing harm trust and brand reputation	Anisah and Wasesa (2025)
Decision-Making Opacity	Delegation of authority to unsupervised systems limits explainability and human oversight	Kumar et al. (2021)
Knowledge Management	Overreliance on AI for high-stakes, rapid decisions based on potentially noisy or biased data	Tong and Tian (2023)

nature of AI models makes it difficult to understand how and why a given classification, prediction or recommendation is generated, creating an appearance of objectivity that can be misleading, especially in high-impact decision-making contexts such as healthcare, financial markets or public policy.

Current scraping and AI technologies do not guarantee the veracity of the extracted content, nor the fairness of the resulting decision-making processes. Entrusting these tools with the task of supporting or automating strategic decisions implies a transfer of authority to the models, without ensuring that there is real human control or verification of the underlying sources. In this sense, trust in AI risks becoming based more on computational power than on the transparency of processes. To address these critical issues, future research will have to invest in the development of explainable AI (XAI) frameworks, capable of making the decision logics of models accessible and interpretable (Raj et al. 2023), as well as integrating with source validation systems and methods for the automatic detection of bias in collected data (Dwivedi et al. 2023). Urgent reflection also concerns the possibility of creating reliable metrics for assessing the "operational truth" of the collected content technically, ethically and socially. Furthermore, it will be crucial to question who decides what is worthy of analysis, what data is systematically excluded, and what worldviews are implicitly encoded in algorithmic architectures.

Another of the most relevant trends in contemporary technological landscape concerns the search for increasingly scalable, adaptive and efficient solutions to deal with the enormous amount of web data generated on a daily basis. In this context, web scraping and AI models are employed to extract, organize and interpret real-time data

Table 13 Future research agenda

Category	Research Gap	Objective	Research Questions
Technical	Scalability and efficiency in processing large volumes of data	Develop algorithms and architectures that can efficiently handle the huge and growing amount of web data in real time	How can scraping and NLP systems be optimized to maintain high performance on ever-growing datasets? Which distributed architectures offer the best trade-off between scalability, latency, and accuracy? How can edge computing be integrated into real-time data collection and analysis workflows?
	Quality and verifiability of data from web scraping	Develop methods to evaluate, filter and validate the veracity and reliability of sources automatically collected from the web	How can mechanisms for evaluating the credibility of sources be integrated into automated scraping processes? Are there strategies to classify collected contents based on their level of information accuracy? How can the "operational truth" of extracted data be measured and communicated in different application contexts?
	Bias amplification in AI models	Detect, mitigate and correct biases in data and NLP models built on web content	What types of biases (gender, cultural, political, etc.) emerge most frequently in datasets derived from web scraping? How do AI models amplify or distort biases already present in the data? What are automatic de-biasing mechanisms that do not compromise model performance?
	Opacity of AI models ("black box")	Improving transparency, interrogability and explainability of decisions taken by combined web scraping and AI models	Which XAI strategies are most effective in making complex neural network-based models interpretable? Which mechanisms can ensure transparency in the transformation of collected data into intelligent outputs? How to formalize a decentralized log system to monitor the data lifecycle in AI-driven contexts?
	Understanding complex language	Improving the ability of NLP models to capture the pragmatic and implicit dimension of human language	Which strategies allow us to systematically detect and correct interpretative errors related to sarcasm, irony or ambiguity? What language monitoring strategies allow us to detect the emergence of new patterns in language? Are there any fine-tuning strategies that make LLMs more sensitive to the pragmatic component without compromising their generality?
Social	Cultural and pragmatic insensitivity of AI models	Explore how choices in data collection, selection and modeling influence the representations produced by models and their social effects	What are the criteria for deciding which data to include/exclude in analyses? What is the impact of cultural specificities in the construction of training datasets? How can we ensure that models are transparent in showing the limitations of their cultural understanding?

Table 13 (continued)

Category	Research Gap	Objective	Research Questions
	Lack of human control in decision-making	Reintroducing human control in automated decision-making processes, through AI auditable and transparent supervision	How can effective human supervision be structured in AI systems that analyze unstructured data? Which tools can help visualize and validate the decision-making paths of AI models before action? Is it possible to formalize roles of "algorithmic responsibility" in automated decision-making flows?
	Lack of informed consent in the collection and use of data via scraping	Define clear, understandable and operational mechanisms to ensure informed consent in the use of data extracted from the web	Which forms of consent can be applied in digital environments with high frequency and communicative informality? Where is the boundary between social analysis and behavioral manipulation? What are the ethical implications of using sentiment analysis to modulate advertising messages or price flows?
Cybersecurity	Differentiated data processing based on their level of sensitivity and context	Design intelligent models capable of adapting their behaviors and decision-making criteria based on the application context and the nature of the data	What characteristics must a model possess to recognize and react to the ethical and regulatory variability of different application contexts? What parameters should guide the access, storage and analysis of sensitive data? What criteria determine the legitimacy of data use in highly sensitive domains?
	Deepfakes, disinformation and multimodal manipulation	Prevent and detect the misuse of AI for the generation of realistic synthetic content for fraudulent or manipulative purposes	How to automatically identify deepfake content generated from scraped data? Which cues (visual, semantic, acoustic) are more effective in distinguishing authentic content from synthetic content? How can content verification systems be integrated into multimodal AI models?
	Digital infrastructure vulnerabilities	Identify and mitigate weaknesses in AI and scraping-based infrastructures that can be exploited for targeted attacks	Which infrastructure configurations expose scraping systems to intrusion or manipulation risks? How can an effective defensive architecture be designed against automated social engineering attacks? What are the main cyber vulnerability risks related to AI advances?
Regulatory	Lack of systemic assessment of the environmental impact of AI and scraping technologies	Integrate environmental sustainability into the design, adoption and evaluation criteria of scraping and AI-based technologies, considering the entire life cycle of the model	What methodologies can be adopted to transparently measure the ecological footprint of an AI system, from scraping to inference? Which international standards could guarantee comparable energy reporting between models and infrastructures? What are the thresholds beyond which the performance gain does not justify the increase in consumption?

Table 13 (continued)

Category	Research Gap	Objective	Research Questions
	Geographic concentration of research and infrastructural inequalities	Balance the distribution of scientific and technical capacity, promoting equitable access to AI development and control tools	What strategies can foster the decentralization of technological and scientific production? What is the role of public institutions in legitimizing and regulating algorithmic infrastructures? How build international consortia for the ethical, equitable and distributed management of scraping and AI infrastructures?
	Need for multidisciplinary governance for the responsible use of AI	Build an integrated regulatory and ethical framework that includes engineering, legal and sociocultural aspects to regulate the use of AI and scraping	Which actors should be involved in the development of binding ethical guidelines for scraping and AI? What is the role of corporate policies in regulating the use of scraping and AI in public or sensitive sectors? What are the solutions to build warning systems for the detection of abuse in the use of AI-generated content?

from increasingly heterogeneous sources for predictive, business or decision-making systems (Bhat and Jain 2024). However, the exponential growth of data and the continuous evolution of natural language pose structural challenges that current models struggle to handle (Khurana et al. 2023).

The rigidity of current models, which often require extensive retraining to adapt to new domains or language changes, limits their usefulness and dynamism. Future research must therefore focus on reliable and adaptive learning models, capable of incorporating new data and language patterns without onerous retraining, exploiting approaches such as few-shot learning (Pragyan et al. 2024), edge computing (Zhang et al. 2022) and distributed processing (Boppana and Sandhya 2023) to maintain real-time responsiveness and security. In parallel, a gap needs to be filled regarding the absence of models designed to be truly sensitive to the cultural and pragmatic variety of human language, going beyond syntactic or semantic correctness. To meet these challenges, technical innovations that see a greater intersection between computer science, linguistics and cognitive sciences will be needed to steer development towards a deeper understanding of human communication in its more subtle and ambiguous aspects.

6.2.2 Cybersecurity insights

Moreover, the unaccountable development of such technologies poses growing risks in the area of cybersecurity and information integrity. For instance, the growing capacity of AI models to analyze and generate multimodal content (i.e. text, images, audio and video) has opened up innovative scenarios but also introduced new risks: from the manipulation of emotions through sentiment analysis to the possibility of producing verisimilar synthetic content (deepfake) used for scams, blackmail or disinformation (Campbell et al. 2022). In this context, the combination of automated scraping of sensitive data and autonomous generation of compromising content

exposes individuals and organizations to new forms of social engineering, in which digital identity is tracked and instrumentalized (Usman et al. 2024).

In fields such as medicine, such scraping and AI are increasingly used to digitize clinical taxonomies (Excoffier et al. 2024) and support diagnoses (Abdullahi et al. 2024). However, the increasing usefulness of these technologies in knowledge management clashes with the highly sensitive nature of the data. Biometric data, for example, are unique identifiers and, if compromised, cannot be 'changed' like a password. Centralizing and automating the processing of highly sensitive data implies guarantees of privacy, informed consent, scientific validation and ethical oversight (El Kababji et al. 2023).

In the field of human resources management, several firms collect and analyze job postings on online platforms to understand skills trends (Dawson et al. 2020) while others develop AI-based solutions to automatically analyze candidates' CVs and make hiring decisions (Ghadekar et al. 2023) without pose significant attention to regulations (Martínez and Fernández 2019).

In highly sensitive sectors such as energy and the military, the ability to collect and analyze large volumes of data from open sources (e.g., exposed industrial dashboards, unsecured technical documentation, satellite image metadata) enables the extraction of critical information about infrastructure, operations and assets with potential consequences related to cyber vulnerability (Knieps 2024) and national security (Lei et al. 2025).

These examples demonstrate that not all data are the same, but their value, sensitivity and possible consequences vary according to context. Despite this, collective awareness of cyber risks related to these technologies remains low. Vulnerability is amplified by the lack of transparency in data collection and processing pipelines, which often centralize highly sensitive information without fully complying with the principles required by regulations such as GDPR (Bodini 2024; Hassani 2024).

To address these threats, specific interdisciplinary studies are needed that combine technical, legal and sociological perspectives to define the boundaries within which it is licit and useful to apply web scraping and AI, clarifying the operating conditions, regulatory limits and forms of liability involved. Future research should prioritize binding ethical guidelines differentiated by sector, early warning systems for abuse of generated content, and "by design" scraping architectures oriented towards security, transparency and respect for digital identity. Without such an effort, there is a real risk that tools designed to maximize efficiency will be applied in areas where the protection of dignity, security and information integrity must take precedence over operational efficiency. The challenge for future research will therefore be to develop contextually sensitive intelligent architectures, flanked by dynamic regulatory frameworks, capable of regulating not only what these technologies can do, but how and why they do it in different domains.

6.2.3 Social insights

The digital transformation has profound social implications that are often ignored in the technical literature.

Firstly, the automation of collective listening through sentiment analysis is based on the presumption that emotions are formalizable, i.e. reducible to labels. But human language is complex, ambivalent, contextual and culturally influenced. Although AI models, especially those specialized in NLP, are advancing rapidly, significant limitations persist in understanding context, idiomatic expressions, irony and sarcasm (Chaudhary and Kakkar 2021), which are fundamental elements for an authentic and non-mechanical interpretation of natural language.

In application scenarios such as customer service automation (Prentice et al. 2020) or reputation analysis (Anisah and Wasesa 2025), the risk is that the algorithmic interpretation of emotions and opinions produces oversimplifications or distortions that lead to erroneous conclusions.

Moreover, when the collection and analysis of emotional data takes place without transparent control, the door is opened to advanced forms of information exploitation, in which collective feelings become a resource to be capitalized upon. The possibility that publicly expressed moods, often in informal and unconscious contexts, are exploited for pricing strategies, opinion manipulation or consensus modulation, raises urgent questions about the boundary between monitoring and surveillance, between prediction and manipulation (Campbell et al. 2022; Shinde et al. 2022).

The web, from being a public space of expression and interaction, is progressively being recoded as a domain functional to the goals of analysis algorithms. The fact that opinions, emotions and behavior expressed in informal environments become the object of algorithmic surveillance raises urgent ethical questions. Who really consents to their emotions expressed in a tweet being used to influence sales strategies? When sentiment analysis tools are fed into automated decision-making processes (e.g., adjusting prices, modulating advertising messages or anticipating market flows) we risk confusing statistical correlations with social truths.

Added to this is the naturalization of algorithmic judgement, whereby classifications and predictions are treated as if they were objective, forgetting that models are trained on noisy, incomplete and often distorted data. In addition to the risk of oversimplifying social complexities, there is also the risk of excluding from decision-making processes all information that is not formalized or computable as forms of local, experiential knowledge.

If the decision-making field moves towards greater algorithmic efficiency while consciously ignoring the context in which the information exists, tools designed to support the understanding of data run the risk of being used to anticipate and direct individual behavior without the providers of that data being informed and aware of it.

What is needed is a broader and more conscious scientific reflection that questions the very conditions of production, validation and legitimization of automated knowledge. Without this critical look, the risk is that collecting data and developing ever more advanced AI models will not help us to understand the world better but only to see what is already compatible with the models.

6.2.4 Regulatory insights

Ethical considerations introduce not only social but also geopolitical implications. The glaring disparities between countries in scientific production revealed by the

results raise major concerns. The concentration of research in a few nations (i.e., China, India, and United States) represents a centralization of knowledge and infrastructural power. In this perspective, technologies for automatic data collection and analysis risk being developed and applied according to opaque logics, oriented more towards surveillance, economic optimization or competitive advantage than towards fair and transparent use. In such a scenario, the least developed countries find themselves in a position of dependence, lacking the resources to build autonomous and sustainable alternatives (Campanile et al. 2024).

To counter this technocratic drift, it is crucial to promote models of open innovation, capable of redistributing cognitive and computational capabilities on a global scale (Zhao and Yang 2023). Companies like Hugging Face, which offers open access to language models, datasets and ML tools while maintaining an active effort towards the democratization of digital technologies, demonstrate that it is possible to build collaborative and transparent digital infrastructures (Taeihagh 2025).

Future research needs interdisciplinary reflection that addresses the cultural, political and normative assumptions of their use. Priorities include the need for independent algorithmic auditing systems, the construction of truly operational forms of informed consent in digital contexts, and the development of democratic governance models of automated knowledge infrastructures. Without these interventions, the use of AI risks consolidating new forms of surveillance and inequality, disguised as technological neutrality and innovation.

Finally, despite the increasing focus on efficiency and automation, the integration of web scraping and AI technologies continues to be misaligned with environmental urgencies. Although these tools offer potential for climate adaptation in agriculture (Nguyen et al. 2024), energy resource optimization (Wang et al. 2025) or traceability of supply chains (Zhu and Vuppalapati 2024), their implementation is not accompanied by a systemic assessment of their direct ecological impact. In particular, the training and execution of large-scale AI models involve high energy consumption, linked both to the computational power required and to the physical location of data centers, which are often powered by non-renewable sources (Patterson et al. 2022). This contradiction results in a glaring gap in the current literature, where environmental sustainability remains marginal to the dominant rhetoric of efficiency and scalability.

Future research must address this disconnect through at least two complementary directions. On the one hand, ‘energy-aware’ scraping methodologies and AI models need to be developed, optimized to reduce computational impact without sacrificing accuracy. On the other, there is a need to promote a systematic assessment of the environmental footprint throughout the entire algorithm life cycle, from data acquisition to inference via training. This also implies greater transparency on the real energy consumption of models and a critical reflection on cost-benefits (e.g. which types of data and analyses really ‘deserve’ the use of computationally intensive technologies), avoiding that the use of AI becomes the default solution to problems that could be solved with less impactful tools.

7 Conclusions

This study investigated the current key research directions in the combined use of advanced AI-based models and web scraping in business contexts. Through a bibliometric analysis, three main thematic clusters (i.e., "Intelligent Web Crawling", "Advanced Sentiment Analysis", and "Digital Social Network") were identified, revealing a landscape dominated by applications focused on automation, predictive analytics, and consumer sentiment interpretation. These findings offer insights into the evolution of technological trends, providing useful knowledge for both academia and management practitioners, while highlighting critical opportunities and challenges in the field.

The analyses conducted in this study offer a valuable academic contribution in several respects. The in-depth exploration of the data sample provides a comprehensive, disciplinary and temporal overview of the research on the combined use of web scraping and AI-based models. The scientific mapping further reveals the centrality of topics such as data mining, machine learning, and NLP by illustrating the transition from early-stage data organization to more sophisticated forms of semantic analysis and business intelligence.

However, the research also highlights significant gaps and critical tensions. The focus on commercial optimization tends to overshadow essential concerns about data privacy, consent, and the socio-political implications of large-scale data mining and algorithmic classification. In particular, the growing use of sentiment analysis and predictive systems raises questions about the potential simplification of human emotions into computational labels and the risk of normalizing algorithmic judgments as objective truths. Furthermore, there is a marked absence of attention to governance models and algorithmic transparency, with the consequence that non-computable data remains excluded from decision-making systems.

Moreover, it exposes the environmental blind spot in current literature, revealing a disconnect between the pursuit of computational performance and the lack of concern for the energy consumption and carbon footprint of large-scale models. Lastly, it brings attention to the geopolitical asymmetries in research and infrastructure, where a concentration of power in a few technologically dominant countries raises concerns about data sovereignty and global inequality. Together, these insights call for a more integrated and ethically grounded theoretical framework that recognizes security, sustainability, and inclusivity as core pillars in the development of web scraping and AI technologies.

Although cybersecurity is mentioned in relation to the combined use of web scraping and AI, the topic appears to be treated in a marginal way. Instead, this study reframes cybersecurity as a structural issue in automated data ecosystems, emphasizing how vulnerabilities in scraping pipelines and the misuse of AI models (e.g., deepfakes, phishing) that pose systemic risks. Moreover, the results expose the environmental blind spot in current literature, revealing a disconnect between the pursuit of computational performance and the lack of concern for the energy consumption and carbon footprint of large-scale models. Lastly, it brings attention to the geopolitical asymmetries in research and infrastructure, where a concentration of power in a few technologically dominant countries raises concerns about data sovereignty and

global inequality. Together, these insights call for a more integrated and ethically grounded theoretical framework that recognizes security, sustainability, and inclusivity as core pillars in the development of web scraping and AI technologies.

The practical implications of this study are therefore twofold. On the one hand, the analysis conducted provides concrete insights for managers and practitioners who wish to exploit AI and web scraping in marketing, sales and strategic decision-making. On the other hand, the study calls for a more critical and interdisciplinary approach to the development of these technologies, integrating technical innovation with ethical, legal, environmental and socio-cultural reflection.

The gaps that emerged from the analysis have been translated into a future research agenda that includes the development of privacy and security-oriented "by design" architectures, the design of energy-sustainable models, the development of explainable AI systems, as well as the definition of effective mechanisms for informed consent and democratic data governance. The aim is to provide a solid basis for future studies that do not only improve the technical performance of the tools, but also contribute to making the technological ecosystem more equitable, transparent and sustainable.

However, the study has some limitations that are important to highlight for a correct contextualization of the results. First, although the bibliometric analysis is based on quantitative data extracted from reliable sources, the interpretation of the thematic clusters and research trajectories inevitably reflects a component of subjectivity on the part of the authors. The choices made in the coding, categorization and critical reading of the results can influence the narrative that emerged, especially in the passages in which interpretative readings were proposed. Second, the highly dynamic nature of the studied field means that some recent, rapidly evolving trends may not yet be fully visible or codified in the bibliometric datasets, making it necessary to continuously update the analyses.

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Declarations

Competing interest The authors have no relevant financial or non-financial interests to disclose.

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Authors and Affiliations

Giuliana Barba[1] · Marianna Lezzi[1] · Mariangela Lazoi[1] · Angelo Corallo[1]

✉ Giuliana Barba
giuliana.barba@unisalento.it

Marianna Lezzi
marianna.lezzi@unisalento.it

Mariangela Lazoi
mariangela.lazoi@unisalento.it

Angelo Corallo
angelo.corallo@unisalento.it

[1] Dipartimento di Ingegneria dell'Innovazione, Università del Salento, Campus Ecotekne, Via per Monteroni, s.n., 73100 Lecce, Italy